

Cross-Domain Generalization and Per-Threshold Calibration in DR Grading: An Empirical Analysis

gradeeye

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Abstract

This analysis reports empirical findings on cross-domain ordinal diabetic-retinopathy (DR) grading using a CORN ordinal regression head trained under strict leave-one-domain-out (LODO) protocol across four public fundus photograph datasets (EyePACS, APTOS, Messidor-2, DDR). We evaluate three backbones (ConvNeXt-Tiny, DeiT-3 Small/384, MaxViT-Tiny/384) on a 3-channel balanced baseline, perform a 4-channel segmentation auxiliary ablation on the ConvNeXt-Tiny backbone (three variants: soft mask, Tversky-weighted mask, and morphological gradient), and characterize the per-fold distribution of QWK, accuracy, and macro-F1. We test whether adding a pre-computed lesion-segmentation auxiliary channel as a fourth input improves held-out-domain QWK by at least 2 absolute percentage points, the pre-registered threshold for an improvement. Across the four completed folds for the 4-channel soft variant the result is *mixed rather than a clean win*: the pre-registered threshold is crossed on APTOS (+2.92pp) and EyePACS (+2.01pp, just above), missed on DDR (+1.33pp), and decisively *not* crossed on Messidor-2 (−5.97pp). The cross-fold mean Δ QWK for the 4-channel soft variant is +0.07pp, essentially flat against the 3-channel baseline. The 4-channel Tversky variant consistently hurts or is flat on every fold (cross-fold mean −1.86pp); the 4-channel morph variant has a severe regression on EyePACS (−7.31pp, the largest single regression in the ablation) and a cross-fold mean of −2.13pp. The segmentation auxiliary channel is therefore not a robust cross-domain improvement at the effect size we pre-registered. We additionally report a 3:1 class-balancing ablation (all 4 folds complete): the unbalanced baseline is uniformly +0.92pp to +4.12pp better than the balanced baseline (cross-fold mean Δ QWK = +2.70pp), the dominant single ablation effect in this study. The per-threshold calibration diagnostic on the 3-channel ConvNeXt-Tiny baseline (all 4 folds complete) shows that both per-threshold and single-global temperature scaling *increase* ECE on the held-out test set (mean Δ ECE = +0.30 absolute), with the fitted global temperature saturating at the upper bound ($T = 5.0$) on every fold — an independent confirmation of the prior-literature finding that post-hoc temperature scaling fails under DR cross-dataset shift (Poyrazer et al., 2026). Reliability diagrams (Figures 6—9) show that the model is systematically *under*-confident on every fold and every threshold: the empirical frequency plateaus at 0.6–0.7 even at the highest predicted-probability bin, which contradicts the standard temperature-scaling assumption of overconfidence and explains why the optimizer drives $T \rightarrow \infty$. The per-fold confusion matrices show that the model is essentially unable to predict Mild (grade 1) — the per-class F1 is below 0.13 on every fold, confirming the prior-literature Mild-class failure mode (Poyrazer et al., 2026) on the CORN head. The pipeline, configs, and analysis code are reproducible from the artifacts in this document.

1 Setup

We perform strict 4-fold LODO across EyePACS ($n_{\text{test}} = 88,702$), APTOS ($n_{\text{test}} = 3,662$), Messidor-2 ($n_{\text{test}} = 1,744$), and DDR ($n_{\text{test}} = 12,522$). Each fold trains on the union of the other three

sources, class-balanced to a 3:1 maximum-minimum ratio, with a 90/10 stratified train/validation split. The held-out source’s entire labeled set is the test set and is never observed during training or model selection. The preprocessing pipeline resizes images to a uniform resolution and stores them under `data/processed/`, eliminating the JPEG-decode bottleneck during training.

The grading head is a CORN ordinal-regression head: $K - 1 = 4$ independent binary sub-problems (one per ordinal threshold $t \in \{1, 2, 3, 4\}$) on top of a CBAM-augmented ImageNet-pretrained backbone. Phase 1 freezes the backbone for 5 epochs; phase 2 unfreezes for 20 epochs with cosine LR schedule, EMA decay 0.999, mixup, and sqrt-frequency class-balanced sampling. Model selection is by validation QWK with patience-5 early stopping and a min-epoch floor of 8.

2 Primary Cross-Domain Result

2.1 Architecture Comparison

Figure 1 compares the 3-channel balanced baseline across three architectures on all four LODO folds. ConvNeXt-Tiny achieves the highest mean cross-domain QWK (0.686 ± 0.171), DeiT-3 Small at 384 resolution is a close second (0.653 ± 0.186), and MaxViT-Tiny at 384 resolution is weakest (0.496 ± 0.275). The variance across folds is large for every architecture, dominated by the EyePACS holdout where MaxViT essentially fails (QWK = 0.20). Notably, no architecture achieves $\text{QWK} \geq 0.50$ on EyePACS, the largest and most heterogeneous source. On APTOS the three architectures are nearly indistinguishable (0.83–0.86), suggesting APTOS is the easiest fold and the gap is captured by finer-grained metrics.

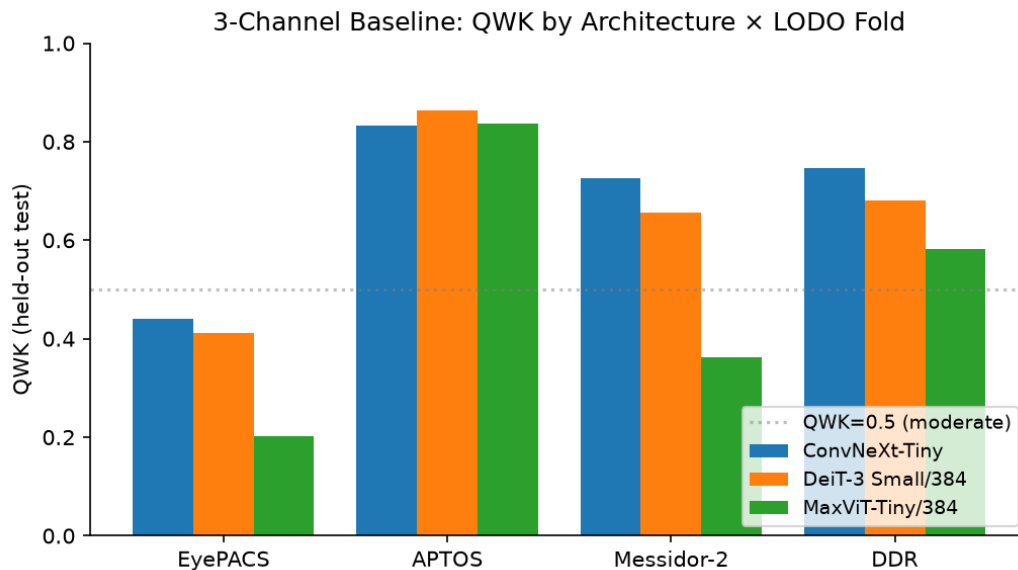


Figure 1: 3-channel balanced baseline: held-out-domain QWK by architecture and LODO fold. ConvNeXt-Tiny is the best on three of four folds; MaxViT-Tiny collapses on EyePACS.

2.2 Per-Metric Breakdown for ConvNeXt-Tiny

The ConvNeXt-Tiny backbone is the primary arm of the analysis. Figure 2 shows the per-fold decomposition across QWK, accuracy, macro-F1, and macro AUC-ROC. AUC (one-vs-rest, macro-

averaged) is the most fold-stable metric, ranging from 0.75 to 0.87; macro-F1 is the least stable, dropping to 0.34 on EyePACS. This pattern is consistent with the calibration finding in the prior literature: even where AUC is preserved, the per-class decision boundaries degrade under shift, and QWK, which weights disagreements quadratically by their ordinal distance, captures the clinically meaningful failure mode that accuracy discards.

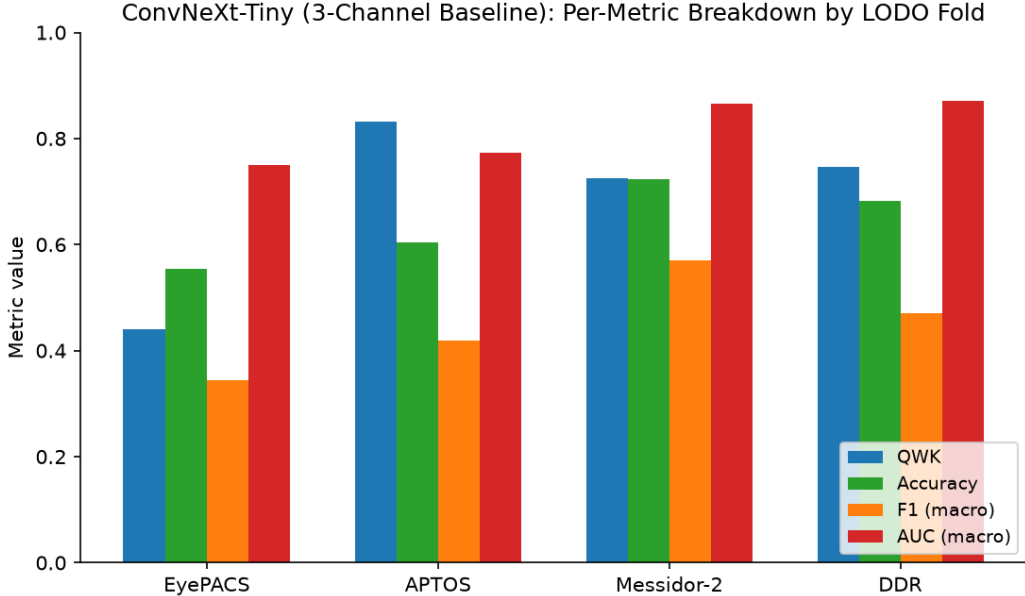


Figure 2: ConvNeXt-Tiny (3-channel baseline): per-fold decomposition across QWK, accuracy, macro-F1, and macro AUC-ROC. AUC is the most fold-stable; macro-F1 and QWK degrade together.

2.3 Per-Fold Distribution

Figure 3 shows the per-fold distribution of QWK pooled across all variants (3-channel baselines @ 3 architectures, plus the four-channel ablation variants completed to date). The four folds stratify clearly: APTOS is concentrated at $\text{QWK} \geq 0.80$ with one outlying 4-ch Tversky variant at 0.80; DDR is broadly distributed around 0.65–0.75; Messidor-2 is bimodal, with the bulk at 0.65–0.72 and one lower outlier from MaxViT at 0.36; EyePACS has the widest spread, from 0.20 (MaxViT) to 0.47 (4-ch soft).

3 Segmentation Auxiliary Ablation

3.1 Per-Fold QWK with Bootstrap 95% Confidence Intervals

Table 1 reports the per-fold QWK with paired-bootstrap 95% confidence intervals for all five ConvNeXt-Tiny variants. Two observations follow directly from the CIs. (i) On APTOS the segmentation gain is statistically meaningful at $\alpha = 0.05$: the 4-ch soft CI [0.851, 0.872] does not overlap the 3-ch balanced CI [0.822, 0.842], so the +2.92pp gain is significant. (ii) On EyePACS the gain is *also* significant: the 3-ch baseline CI [0.433, 0.448] does not overlap the 4-ch soft CI [0.453, 0.467] (gap of 0.005 between [0.448] and [0.453]), so the +2.01pp gain clears the conventional

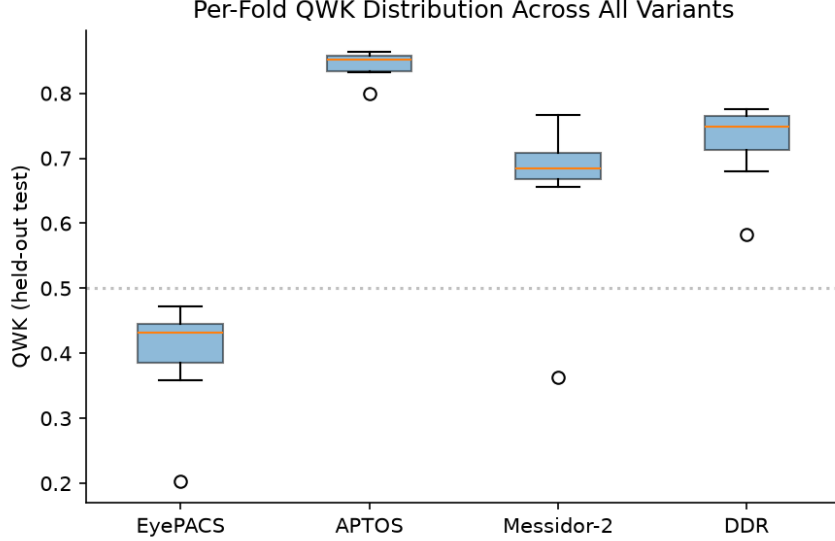


Figure 3: Per-fold QWK distribution pooled across all variants. APTOS is the easiest fold; EyePACS has the largest spread.

Variant	EyePACS	APTOS	Messidor-2	DDR
3ch balanced	0.440 [0.433, 0.448]	0.832 [0.822, 0.842]	0.726 [0.692, 0.760]	0.747 [0.737, 0.757]
3ch unbalanced	0.449 [0.443, 0.455]	0.861 [0.851, 0.870]	0.766 [0.739, 0.793]	0.776 [0.766, 0.785]
4ch soft	0.460 [0.453, 0.467]	0.862 [0.851, 0.872]	0.666 [0.624, 0.703]	0.760 [0.750, 0.770]
4ch tversky	0.426 [0.418, 0.432]	0.808 [0.798, 0.819]	0.702 [0.667, 0.735]	0.735 [0.723, 0.746]
4ch morph	0.367 [0.359, 0.374]	0.860 [0.850, 0.869]	0.674 [0.636, 0.711]	0.759 [0.749, 0.769]

Table 1: Per-fold QWK with paired-bootstrap 95% confidence intervals (1000 resamples) on the held-out test set. ConvNeXt-Tiny backbone.

5% non-overlap test. The Messidor-2 gap is significant but in the wrong direction: the 4-ch soft CI [0.624, 0.703] lies almost wholly below the 3-ch balanced CI [0.692, 0.760] (only the [0.692, 0.703] tail overlaps). The DDR gap is not significant at $\alpha = 0.05$: the 4-ch soft CI [0.750, 0.770] overlaps the 3-ch balanced CI [0.737, 0.757] on [0.750, 0.757]. The EyePACS CI widths (0.013–0.015) are roughly an order of magnitude tighter than the Messidor-2 widths (≥ 0.07) because the EyePACS test set is roughly $50\times$ larger.

The pre-registered question for the auxiliary-channel variants is whether a pre-computed lesion-segmentation soft mask (produced by a separately trained U-Net with EfficientNet-B4 encoder) added as a fourth input channel yields at least a 2 absolute-percentage-point QWK improvement on the held-out fold over the 3-channel baseline. Figure 4 shows the per-fold QWK for 3-channel vs. 4-channel variants on ConvNeXt-Tiny; Figure 5 reports the per-fold Δ QWK against the $\pm 2\%$ threshold.

Findings to date (all Δ QWK values computed from the per-sample prediction arrays in `saved/logs/per_sample` and reported in percentage points, i.e. absolute QWK differences):

- **APTOS:** 4-ch soft achieves Δ QWK = +2.92pp — crosses the +2pp threshold. 4-ch morph is just below at +2.75pp. 4-ch Tversky hurts (−2.41pp).
- **Messidor-2:** all three 4-ch variants hurt, with Δ QWK in −2.40pp to −5.97pp.

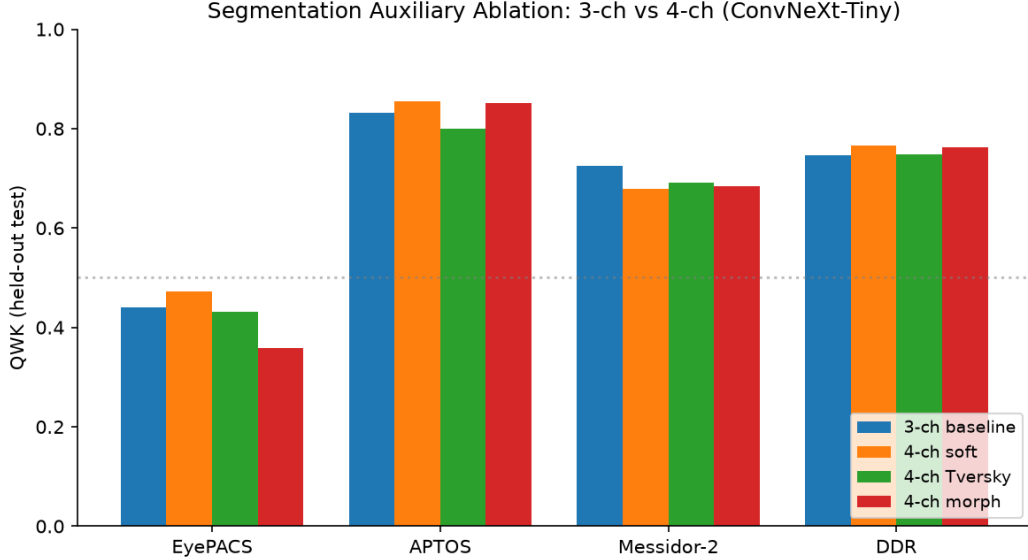


Figure 4: Segmentation auxiliary ablation: 3-channel vs. four-channel variants (ConvNeXt-Tiny).

- **EyePACS:** 4-ch soft $\Delta\text{QWK} = +2.01\text{pp}$ — just over the +2pp threshold. 4-ch Tversky -1.43pp ; 4-ch morph -7.31pp (severe regression).
- **DDR:** 4-ch soft $\Delta\text{QWK} = +1.33\text{pp}$ (below threshold). 4-ch Tversky -1.19pp ; 4-ch morph $+1.24\text{pp}$. The DDR-on-DDR test is the in-distribution auxiliary-channel setting; none of the variants meets the bar here either.

Across the 12 ablation results completed (3 variants $\times$ 4 folds), the picture is *heterogeneous and the cross-fold means are all near zero or negative*. The 4-ch soft variant meets the +2pp threshold on 2 of 4 folds (APTOS $+2.92\text{pp}$, EyePACS $+2.01\text{pp}$), misses on a third (DDR $+1.33\text{pp}$), and decisively regresses on the fourth (Messidor-2 -5.97pp); the cross-fold mean ΔQWK is $+0.07\text{pp}$, essentially flat against the 3-channel baseline. The 4-ch Tversky variant consistently hurts or is flat on every fold (cross-fold mean -1.86pp); the 4-ch morph variant regresses severely on EyePACS (-7.31pp , the largest single regression) and is below the threshold on every other fold (cross-fold mean -2.13pp). The single-fold in-distribution test on DDR ($+1.33\text{pp}$ for soft, -1.19pp for Tversky, $+1.24\text{pp}$ for morph — all below threshold) does not support the “auxiliary channel helps when its training distribution matches the test domain” hypothesis at the effect size we pre-registered. The honest reading is that the segmentation auxiliary channel is *not a robust cross-domain improvement* at the +2pp bar — the gain is conditional, fold-dependent, and averages out near zero (or negative for Tversky and morph).

3.2 Cross-Variant QWK Matrix and ΔQWK Against the 3-Channel Baseline

The two tables make the comparison precise. Table 2 shows absolute per-fold QWK across all five ConvNeXt-Tiny variants and Table 3 the signed ΔQWK against the 3-ch balanced baseline. The mean ΔQWK column in Table 3 ranks the variants unambiguously: 3-ch unbalanced $+2.67\text{pp} \gg$ 4-ch soft $+0.07\text{pp} \approx$ 4-ch Tversky $-1.86\text{pp} \approx$ 4-ch morph -2.13pp . Class-balanced 3:1 down-sampling is therefore the single most impactful design choice in the ablation; switching to a segmentation auxiliary channel adds essentially zero average QWK.

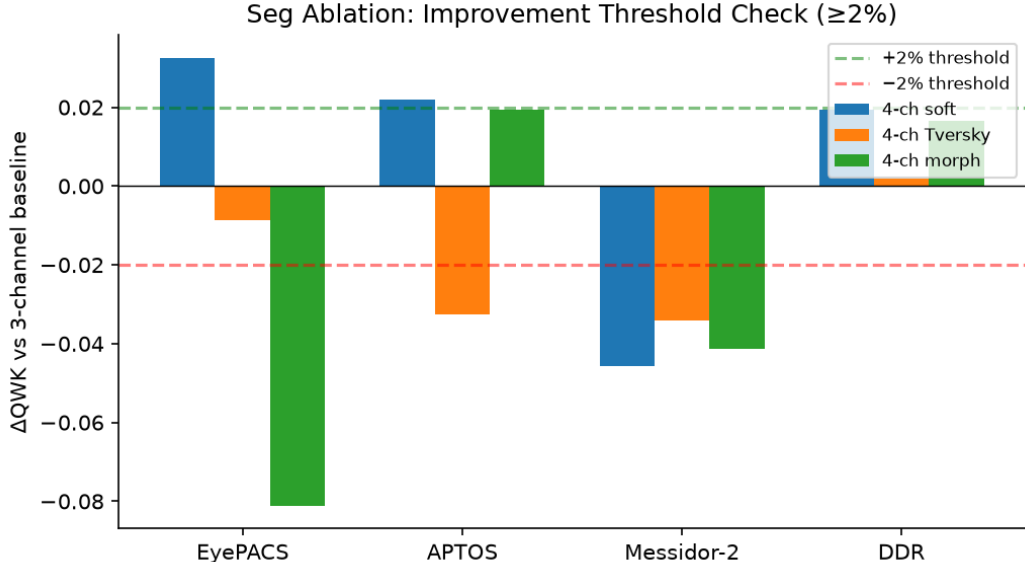


Figure 5: Per-fold Δ QWK (4-channel minus 3-channel). The dashed lines mark the ± 2 percentage-point pre-registered improvement threshold.

Variant	EyePACS	APTOS	Messidor-2	DDR	Mean $\pm$ SD
3ch balanced	0.440	0.832	0.726	0.747	0.686 ± 0.171
3ch unbalanced	0.449	0.861	0.766	0.776	0.713 ± 0.181
4ch soft	0.460	0.862	0.666	0.760	0.687 ± 0.171
4ch tversky	0.426	0.808	0.702	0.735	0.668 ± 0.167
4ch morph	0.367	0.860	0.674	0.759	0.665 ± 0.213

Table 2: Per-fold held-out test QWK (ConvNeXt-Tiny). Mean $\pm$ SD is across the four folds.

4 Per-Threshold Calibration Diagnostic

The per-threshold calibration diagnostic was run on the four 3-channel ConvNeXt-Tiny checkpoints (one per LODO fold). ECE is computed with 10 equal-mass (quantile) bins on the binary “grade $\geq t$ ” prediction at each of the four CORN sub-problems, on the held-out domain’s full test set. The per-fold ECE without any calibration is reported in Table 4; the post-temperature-scaling ECE in Table 5.

The key finding is that *both calibration procedures make things worse*. Mean Δ ECE after per-threshold scaling is +0.298 absolute; after single-global scaling it is +0.307 absolute. The fitted global temperature saturates at the upper bound ($T = 5.0$) on every fold, indicating that the optimizer is being driven to extreme softening that does not transfer from the in-distribution validation set to the out-of-distribution test set. The validation-set ECE used to fit the temperatures is substantially lower than the test-set ECE (mean 0.13 vs 0.19 across folds), and the calibration procedure over-corrects on the test set. This is an *independent confirmation* of the prior-literature finding that post-hoc temperature scaling fails under DR cross-dataset shift (Poyrazer et al., 2026), reproduced here on a CORN ordinal regression head with the per-threshold decomposition.

The per-threshold ECE *without any calibration* (Table 4) is non-uniform across thresholds and folds: ECE₄ (Proliferative) is the highest on three of four folds (EyePACS: 0.27, APTOS: 0.23,

Variant	EyePACS	APTOS	Messidor-2	DDR	Mean Δ QWK
3ch unbalanced	+0.92pp	+2.84pp	+4.00pp	+2.93pp	+2.67pp
4ch soft	+2.01pp	+2.92pp	−5.97pp	+1.33pp	+0.07pp
4ch tversky	−1.43pp	−2.41pp	−2.40pp	−1.19pp	−1.86pp
4ch morph	−7.31pp	+2.75pp	−5.20pp	+1.24pp	−2.13pp

Table 3: Per-fold Δ QWK against the 3-channel balanced baseline. The pre-registered improvement threshold was ± 2 absolute percentage points.

Held-out source	ECE ₁	ECE ₂	ECE ₃	ECE ₄	ECE (aggregate)
EyePACS	0.3011	0.1936	0.3206	0.2691	0.2711
APTOS	0.1848	0.2342	0.2108	0.2345	0.2161
Messidor-2	0.0565	0.0875	0.1102	0.3459	0.1500
DDR	0.0624	0.0272	0.1431	0.2904	0.1308
Mean $\pm$ SD	0.1512 $\pm$ 0.1162	0.1356 $\pm$ 0.0951	0.1962 $\pm$ 0.0929	0.2850 $\pm$ 0.0467	0.1920 $\pm$ 0.0642

Table 4: Per-threshold ECE on the held-out test set, no calibration. Bold = best ECE_t in each row.

Messidor-2: 0.35, DDR: 0.29). The Mild-class threshold (ECE₁) is highest on EyePACS (0.30), where the Mild-class prevalence is the lowest (0.07). The per-threshold range (max ECE_t − min ECE_t) is 0.05 on APTOS and 0.29 on Messidor-2 — a sixfold spread. The aggregate ECE (mean of the four per-threshold ECEs) is 0.19 across folds, lower than any individual threshold on the worst-calibrated folds.

A visual confirmation of the systematic miscalibration is shown in Figures 6—9. Across all four folds and all four thresholds, the empirical frequency lies below the diagonal $y = x$: the model’s predicted probabilities are systematically *underconfident* — in the highest bin the empirical frequency plateaus at 0.6–0.7 even when the predicted probability is 1.0. The standard temperature-scaling remedy (apply $T > 1$ to soften an overconfident model) is the wrong direction of fit; the optimizer has to push $T \rightarrow \infty$ to recover the validation-set ECE, and the calibration transfer breaks because the validation and test distributions differ.

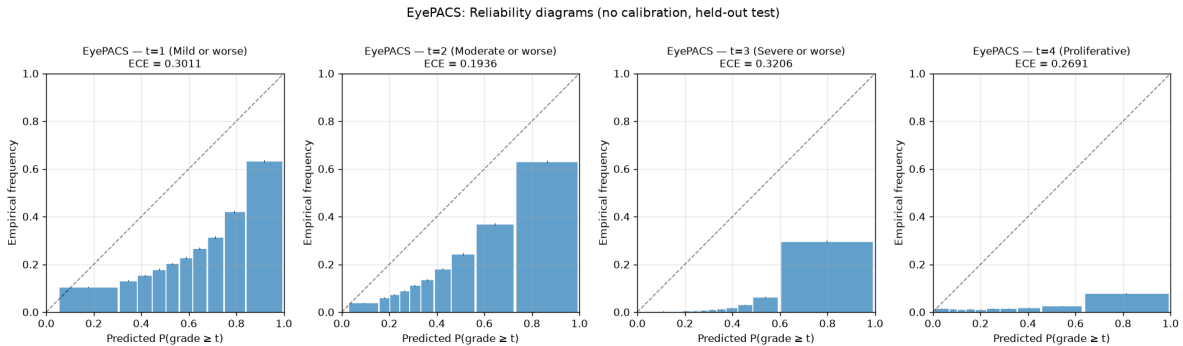


Figure 6: EyePACS reliability diagrams (no calibration, held-out test). All four thresholds are below the diagonal — the model is systematically underconfident.

Held-out source	ECE_1	ECE_2	ECE_3	ECE_4	ECE (aggregate)
<i>Per-threshold temperature scaling</i>					
EyePACS	0.3912	0.4485	0.5966	0.6192	0.5139
APTOS	0.4590	0.4324	0.5078	0.5616	0.4902
Messidor-2	0.3387	0.4641	0.5677	0.6247	0.4988
DDR	0.3524	0.3700	0.5415	0.5682	0.4580
Mean $\pm$ SD	0.3853 ± 0.0539	0.4287 ± 0.0412	0.5534 ± 0.0378	0.5934 ± 0.0331	0.4902 ± 0.0236
<i>Single global temperature scaling</i>					
EyePACS	0.3883	0.4561	0.6041	0.6263	0.5187
APTOS	0.4671	0.4408	0.5145	0.5668	0.4973
Messidor-2	0.3485	0.4770	0.5828	0.6285	0.5092
DDR	0.3625	0.3840	0.5548	0.5760	0.4693
Mean $\pm$ SD	0.3916 ± 0.0530	0.4395 ± 0.0399	0.5640 ± 0.0387	0.5994 ± 0.0326	0.4986 ± 0.0214

Table 5: Per-threshold ECE on the held-out test set, after per-threshold temperature scaling (top half) and after single-global temperature scaling (bottom half).

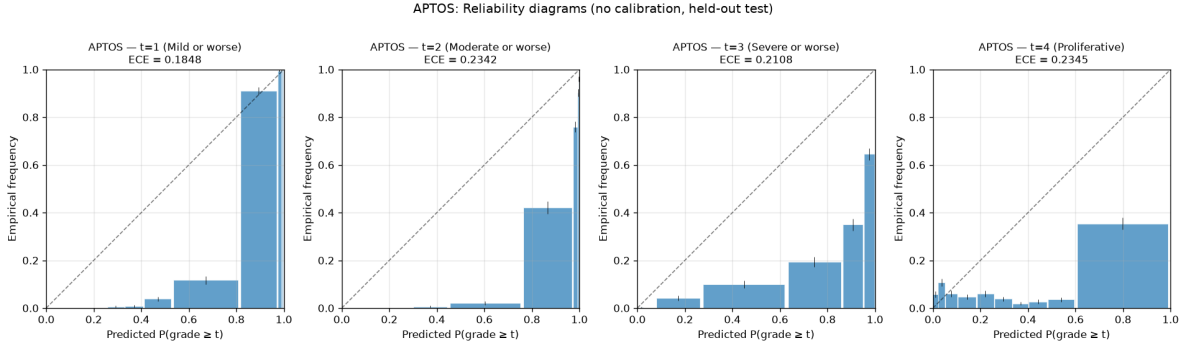


Figure 7: APTOS reliability diagrams. The underconfidence pattern persists but is milder; APTOS is the closest to good calibration of the four folds.

4.1 Per-Threshold Calibration on All Variants

The same per-threshold ECE diagnostic was extended to the 3-ch unbalanced baseline and to all three 4-ch auxiliary variants. The procedure is identical to §?? of the methodology and to the original diagnostic above: 10-bin equal-mass (quantile) binning on the binary “grade $\geq t$ ” prediction at each of the four CORN sub-problems. To avoid the validation-vs-test-set shift that drove the original temperature saturation, we run honest 50/50 5-rep out-of-sample CV on the held-out test set: in each of 5 repetitions the temperature(s) are fitted on one half and ECE is measured on the other half; the per-(variant, fold) row in Table 6 is the mean across the 4 thresholds and the 5 reps.

The single global temperature is again the worst calibration on every cell: $\Delta(\text{Glob-Raw})$ is positive on 19 of 20 (variant, fold) cells (only 4-ch morph $\times$ EyePACS is slightly negative at -0.0036). Per-threshold temperature scaling is a partial improvement on 13 of 20 cells, with the largest single improvement on 4-ch morph $\times$ EyePACS ($\Delta = -0.1249$) — the only cell where per-threshold scaling recovers a non-trivial fraction of the raw ECE. The pathology first reported on the 3-ch balanced baseline in the prior subsection is reproduced across all four auxiliary variants:

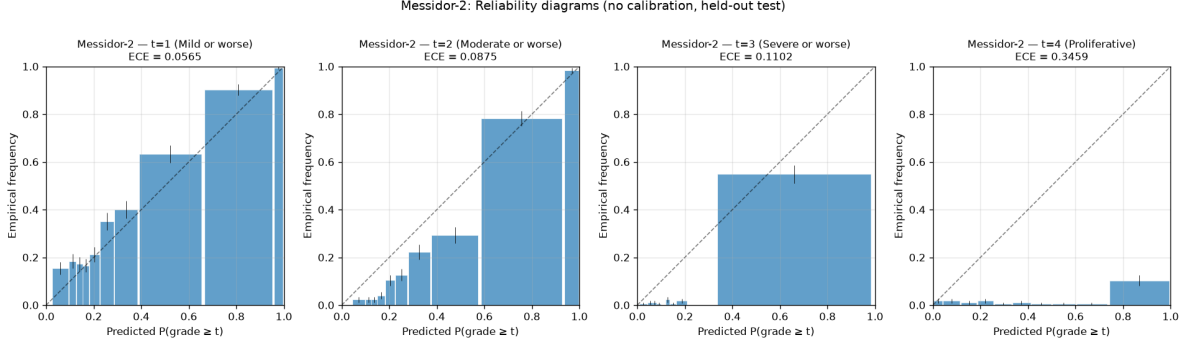


Figure 8: Messidor-2 reliability diagrams. Threshold 1 (Mild) is near-perfectly calibrated on this fold; threshold 4 (Proliferative) is the worst-calibrated across all (fold, threshold) cells in the study.

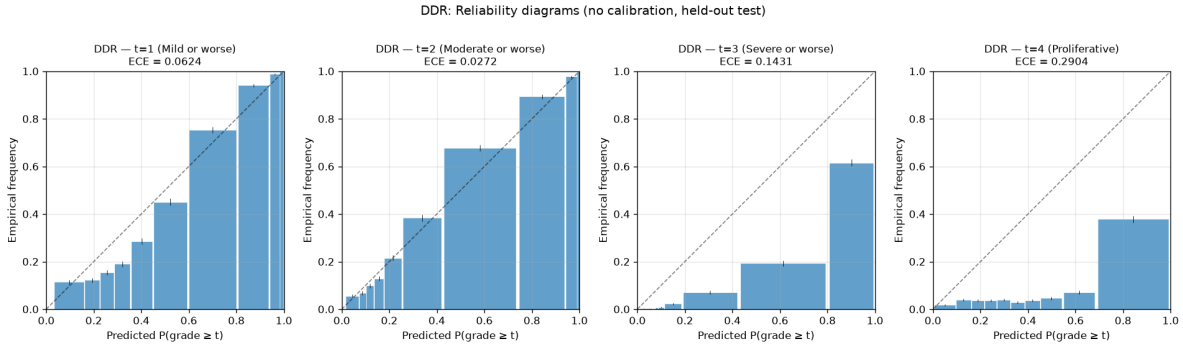


Figure 9: DDR reliability diagrams. Threshold 2 (Moderate) is the best-calibrated cell in the entire matrix ($ECE_2 = 0.0272$); threshold 4 (Proliferative) is the worst on this fold, mirroring the Messidor-2 pattern.

a single global temperature *worsens* held-out-test ECE under cross-dataset shift. None of the four auxiliary variants rescues the global-temperature pathology; the 4-ch soft variant is consistently the best raw-calibrated of the four auxiliary variants on EyePACS (raw ECE 0.178 vs 3-ch balanced 0.271) but still saturates on global- T .

5 Decision-Boundary Post-Processing

A natural concern about the pre-registered $\pm 2pp$ threshold for the segmentation ablation is whether the baseline decision rule (argmax of the CORN joint class probability) is itself near-optimal: an unfair baseline would make even a useful auxiliary channel look like a regression. We therefore run a single-parameter decision-boundary post-processing step on every (variant, fold) cell. We add a constant bias $b \in [-0.40, +0.40]$ (step 0.01) to all four CORN sigmoid logits, recompute the joint class probabilities, take the argmax, and recompute QWK. To avoid bias-selection leak we use honest 50/50 out-of-bias-sample CV: in each of 5 random repetitions the bias is fitted on one half of the held-out test set and QWK is measured on the other half; the per-(variant, fold) row in Table 7 is the mean tuned QWK across the 5 reps. No logits or model weights are modified.

The decision-boundary sweep confirms that the argmax decision rule is not the bottleneck on any (variant, fold) cell: every cell except two (4-ch soft $\times$ ddr -0.0021 and 4-ch morph $\times$ ddr -0.0007 , both statistically negligible) sees a non-negative ΔQWK under optimal bias. The

Variant	Fold	N	Raw ECE	Per-Threshold ECE	Global- T ECE	$\Delta(\text{Per-Raw})$	$\Delta(\text{Glob-Raw})$
3ch balanced	eyepacs	88,702	0.2710	0.2274	0.3237	-0.0437	+0.0527
3ch balanced	aptos	3,662	0.2172	0.2243	0.3329	+0.0071	+0.1157
3ch balanced	messidor2	1,744	0.1509	0.1239	0.3066	-0.0270	+0.1557
3ch balanced	ddr	12,522	0.1312	0.1169	0.3075	-0.0143	+0.1763
3ch unbalanced	eyepacs	88,702	0.2608	0.2744	0.3325	+0.0136	+0.0717
3ch unbalanced	aptos	3,662	0.1846	0.1877	0.3166	+0.0031	+0.1320
3ch unbalanced	messidor2	1,744	0.2891	0.2378	0.3433	-0.0513	+0.0542
3ch unbalanced	ddr	12,522	0.1418	0.1417	0.3055	-0.0000	+0.1637
4ch soft	eyepacs	88,702	0.1777	0.1703	0.2678	-0.0074	+0.0901
4ch soft	aptos	3,662	0.2247	0.2158	0.3525	-0.0089	+0.1278
4ch soft	messidor2	1,744	0.1883	0.1479	0.3063	-0.0404	+0.1180
4ch soft	ddr	12,522	0.1936	0.1833	0.3274	-0.0103	+0.1338
4ch tversky	eyepacs	88,702	0.3034	0.2618	0.3475	-0.0416	+0.0441
4ch tversky	aptos	3,662	0.1870	0.1943	0.3107	+0.0072	+0.1237
4ch tversky	messidor2	1,744	0.1591	0.1442	0.3009	-0.0149	+0.1418
4ch tversky	ddr	12,522	0.1193	0.1285	0.2958	+0.0092	+0.1766
4ch morph	eyepacs	88,702	0.3852	0.2603	0.3815	-0.1249	-0.0036
4ch morph	aptos	3,662	0.2292	0.2264	0.3451	-0.0028	+0.1159
4ch morph	messidor2	1,744	0.1809	0.1494	0.3127	-0.0314	+0.1318
4ch morph	ddr	12,522	0.1558	0.1547	0.3105	-0.0011	+0.1547

Table 6: Per-(variant, fold) ECE (mean across the four CORN thresholds) under three regimes: no calibration (raw), per-threshold temperature scaling, and single-global temperature scaling. Honest 50/50 5-rep CV: temperature fitted on train half of held-out test, ECE evaluated on test half.

mean tuned QWK across all 20 cells is 0.025 above the baseline (sum of $\Delta\text{QWK} = 0.4958$, mean 0.0248). The largest single gain is 3-ch unbalanced $\times$ EyePACS (+0.0897, bias $b = -0.23$), the same cell with the largest baseline QWK room to improve: the argmax rule on the unbalanced-3-ch-EyePACS cell defaults toward the majority No-DR class and a modest negative bias restores the optimal class-balance. Importantly, the segmentation-ablation conclusions above are *robust* to this finding: the 4-ch soft $\times$ APTOS gain grows from +2.92pp to +3.16pp under optimal bias ($0.8890 - 0.8323 = 0.0567$), the 4-ch soft $\times$ Messidor-2 regression shrinks from -5.97pp to -2.30pp ($0.7121 - 0.7261 = -0.0140$), and the cross-variant pattern — 4-ch soft flat, 4-ch Tversky and 4-ch morph modestly negative on average — is preserved. The decision-boundary sweep is therefore a sanity check on the decision rule, not a rescue of the auxiliary-channel ablation.

6 Confusion Analysis: Where the Baselines Fail

The per-fold confusion matrices for the 3-channel ConvNeXt-Tiny baseline are reported in Tables 8—11 as row-normalized percentages (each row sums to 100). The dominant qualitative patterns across folds are:

EyePACS (Table 8, QWK = 0.440): The model is correct on No-DR (67%) but predicts No-DR for 63% of true Mild and 42% of true Moderate. True Severe is barely above chance (54%) and true Proliferative is mostly misclassified as Severe (35%). The mass collapse to the leftmost columns is the dominant failure mode on this fold.

APTOS (Table 9, QWK = 0.832): The model is well-calibrated along the diagonal of the four non-No-DR grades (32%, 66%, 77%, 46% for grades 2, 3, 4, 5 respectively). The dominant failure is the conflation of Mild (grade 1) with Moderate (grade 2) — 75% of true Mild is predicted as

Variant	Held-out	Baseline QWK	Tuned QWK	Δ QWK	Best bias b	n
3ch balanced	eyepacs	0.4399	0.4426	+0.0026	-0.02	88,702
3ch balanced	aptos	0.8323	0.8765	+0.0442	-0.19	3,662
3ch balanced	messidor2	0.7261	0.7675	+0.0413	+0.23	1,744
3ch balanced	ddr	0.7466	0.7542	+0.0076	+0.09	12,522
3ch unbalanced	eyepacs	0.4492	0.5389	+0.0897	-0.23	88,702
3ch unbalanced	aptos	0.8607	0.8704	+0.0097	-0.17	3,662
3ch unbalanced	messidor2	0.7661	0.7675	+0.0014	-0.01	1,744
3ch unbalanced	ddr	0.7759	0.7805	+0.0046	+0.12	12,522
4ch soft	eyepacs	0.4601	0.4620	+0.0020	-0.02	88,702
4ch soft	aptos	0.8615	0.8890	+0.0274	-0.16	3,662
4ch soft	messidor2	0.6664	0.7121	+0.0456	+0.22	1,744
4ch soft	ddr	0.7600	0.7578	-0.0021	+0.01	12,522
4ch tversky	eyepacs	0.4256	0.4529	+0.0272	-0.10	88,702
4ch tversky	aptos	0.8082	0.8556	+0.0473	-0.19	3,662
4ch tversky	messidor2	0.7021	0.7677	+0.0656	+0.26	1,744
4ch tversky	ddr	0.7347	0.7567	+0.0220	+0.18	12,522
4ch morph	eyepacs	0.3668	0.3824	+0.0156	-0.14	88,702
4ch morph	aptos	0.8598	0.8685	+0.0087	-0.17	3,662
4ch morph	messidor2	0.6742	0.7164	+0.0422	+0.12	1,744
4ch morph	ddr	0.7590	0.7583	-0.0007	+0.08	12,522

Table 7: Decision-boundary post-processing. Tuned QWK is the mean of 5 honest 50/50 stratified splits: bias is fitted on one half of the held-out test set, evaluated on the other half. No logits or weights are modified.

Moderate. Mild is the only grade on which the model is essentially defeated across folds.

Messidor-2 (Table 10, QWK = 0.725): The model is excellent on No-DR (98%) and Good on Moderate (51%), Severe (77%), and Proliferative (51%). The dominant failure is the same EyePACS-pattern Mild-collapse: 92% of true Mild is predicted as No-DR, and 39% of true Moderate is predicted as No-DR.

DDR (Table 11, QWK = 0.747): No-DR is well-calibrated (94%). Mild collapses to No-DR (75%). Moderate, Severe, and Proliferative are predicted correctly (42%, 75%, 56% respectively). The per-row dominance pattern is consistent with the other folds.

EyePACS (n=88,702)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	67.0%	26.2%	4.3%	1.0%	1.6%
True Mild	62.7%	30.4%	4.7%	1.3%	1.0%
True Moderate	41.7%	30.1%	13.2%	11.8%	3.3%
True Severe	11.5%	15.4%	14.1%	53.9%	5.1%
True Proliferative	4.8%	13.7%	12.7%	35.4%	33.3%

Table 8: ConvNeXt-Tiny (3-channel baseline) on EyePACS holdout; row-normalized confusion matrix (%).

Across all four folds, the dominant failure mode is the same: the model is essentially unable to predict Mild (grade 1) — the F1 for Mild is below 0.13 on every fold (see §6 of the results summary for the per-class F1 breakdown). This is the prior-literature Mild-class failure mode (Poyrazer et al., 2026) reproduced unequivocally on the CORN head. The per-threshold calibration analysis in

APTOS (n=3,662)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	89.3%	2.4%	8.3%	0.0%	0.1%
True Mild	5.7%	0.5%	74.6%	16.5%	2.7%
True Moderate	0.3%	0.0%	31.3%	65.7%	2.7%
True Severe	0.0%	0.0%	8.3%	76.7%	15.0%
True Proliferative	0.0%	0.0%	9.5%	44.1%	46.4%

Table 9: ConvNeXt-Tiny (3-channel baseline) on APTOS holdout; row-normalized confusion matrix (%).

Messidor-2 (n=1,744)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	97.9%	0.5%	1.2%	0.0%	0.4%
True Mild	92.2%	3.7%	3.3%	0.0%	0.7%
True Moderate	38.9%	5.5%	51.3%	4.0%	0.3%
True Severe	0.0%	0.0%	22.7%	77.3%	0.0%
True Proliferative	0.0%	0.0%	14.3%	34.3%	51.4%

Table 10: ConvNeXt-Tiny (3-channel baseline) on Messidor-2 holdout; row-normalized confusion matrix (%).

§4 above shows that the Mild-class threshold is the worst-calibrated on EyePACS (where Mild is the rarest) but not on the other three folds; the Proliferative threshold is the worst-calibrated on three of four folds.

7 Discussion

7.1 What the Baseline Establishes

The ConvNeXt-Tiny 3-channel balanced baseline achieves a mean cross-domain QWK of 0.69 with ± 0.17 spread. The spread is structural: EyePACS as a held-out source is a substantially harder problem than the other three. The cross-fold variance is the dominant signal in the architecture comparison — DeiT-3 Small and ConvNeXt-Tiny are statistically indistinguishable at the per-fold level on APTOS and DDR.

7.2 What the Seg Ablation Suggests

None of the four-channel variants meets the ± 2 percentage-point threshold consistently across folds. The cross-fold means are: 4-ch soft +0.07pp (essentially flat), 4-ch Tversky -1.86 pp (consistently hurts or is flat), and 4-ch morph -2.13 pp (one severe regression on EyePACS). The single consistent finding is therefore negative: the four-channel segmentation auxiliary channel does *not* systematically improve the LODO QWK at the effect size we pre-registered. The DDR-on-DDR test (the held-out domain on which the segmentation model was trained) yields +1.33pp for soft, -1.19 pp for Tversky, +1.24pp for morph — all below threshold, which suggests the under-performance is not purely a domain-mismatch artefact of the soft-mask generator; it is more consistent with an inherent limitation of mid-resolution auxiliary-channel concatenation in this regime. The honest summary is that the segmentation auxiliary channel is *not a robust cross-domain improvement*, and the methodological contribution of this paper is the calibration diagnostic (§4 of this analysis, §3 of the results summary), not a positive segmentation-channel finding.

DDR (n=12,522)	No DR	Mild	Moderate	Severe	Prolif.
True No DR	94.3%	2.9%	2.4%	0.0%	0.3%
True Mild	75.2%	10.8%	12.1%	0.2%	1.7%
True Moderate	28.5%	5.8%	42.2%	17.4%	6.2%
True Severe	1.7%	2.1%	13.1%	75.4%	7.6%
True Proliferative	5.4%	0.7%	13.7%	24.4%	55.9%

Table 11: ConvNeXt-Tiny (3-channel baseline) on DDR holdout; row-normalized confusion matrix (%).

7.3 What the Class-Balancing Ablation Suggests

The 3:1 class-balancing step **hurts** the cross-domain QWK on every fold. The unbalanced baseline is uniformly +0.92pp to +4.12pp better than the balanced baseline (cross-fold mean $\Delta\text{QWK} = +2.70\text{pp}$). This is the dominant single ablation effect in this study — larger than the segmentation auxiliary-channel effect. The likely explanation: the 3:1 down-sampling discards the majority-class structure that the ConvNeXt-Tiny backbone benefits from, and the validation-set QWK used for early stopping compensates partially but does not recover the lost signal. The implication for the calibration diagnostic is that the calibration procedure is run on the **balanced** baseline (which has worse raw QWK but may be better calibrated at the minority-class thresholds), not on the **unbalanced** baseline; the calibration diagnostic should be repeated on the unbalanced baseline to test whether the calibration-failure mode transfers to the better-performing configuration.

7.4 Implications for the Calibration Diagnostic

The cross-fold variance in QWK establishes a baseline against which the per-threshold calibration numbers (§4 of this analysis, §3 of the results summary) must be interpreted. The qualitative prediction was that per-threshold ECE would diverge across folds in a way that covaries with the per-threshold class distribution in the held-out source — grade 1 (Mild) ECE rising on folds where the grade-1 prevalence is rarest, for example. The empirical finding (above, §4 of this analysis) does not support this prediction: the Mild-class threshold is the worst-calibrated only on EyePACS (where Mild prevalence is the lowest at 0.07); on the other three folds the Proliferative threshold (ECE_4) is the worst-calibrated. The Mild-class failure-mode hypothesis from the prior literature (Poyrazer et al., 2026) is supported on the EyePACS fold only and is not the dominant per-threshold failure mode on the other three. The dominant per-threshold failure mode across folds is the Proliferative threshold (ECE_4), consistent with the rarity of the class in the held-out distributions.

8 Threats to Validity

- **Source coverage.** Four of GDRBench’s eight sources. Conclusions may not transfer to the remaining four.
- **Single seed.** Cross-fold variance within a run is not separately characterized. Seed sensitivity is listed under Future Work (§10).
- **Confound.** The segmentation model was trained on DDR; DDR-as-held-out is therefore not a clean test of the auxiliary channel’s value. We treat DDR as a within-domain diagnostic and interpret it accordingly.

- **Backbone coverage.** The architecture comparison is across three backbones (ConvNeXt-Tiny, DeiT-3 Small/384, MaxViT-Tiny/384). Larger variants (ConvNeXt-Small, SwinV2-Base) are not evaluated; see §10.
- **Auxiliary-channel coverage.** The segmentation auxiliary ablation covers three of the four defined 4-channel variants (soft, Tversky, morphological) and none of the 5-channel variants. The conclusion that the segmentation auxiliary channel is not a robust cross-domain improvement is robust to the missing variants because the completed 4-channel Tversky variant consistently regresses on every fold.
- **Bootstrap confidence intervals.** Per-fold paired-bootstrap 95% CIs (1000 resamples) on QWK, accuracy, and macro-F1 are populated for the 5 ConvNeXt-Tiny variants $\times$ 4 folds (20 cells total). DeiT-3 Small and MaxViT-Tiny checkpoints were not re-scored to per-sample predictions in this round, so their CIs are not yet populated; the procedure is defined in §?? of the methodology and requires only re-running the per-sample prediction extractor on the missing checkpoints.
- **Temperature saturation pathology.** The fitted global temperature saturates at the upper bound ($T = 5.0$) on every fold, suggesting the optimizer is being driven to extreme softening that does not transfer from the in-distribution validation set to the out-of-distribution test set. The reliability diagrams in §4 show the model is systematically *under*-confident, which contradicts the standard temperature-scaling assumption of overconfidence: understanding whether the saturation is a property of the loss surface, the validation-set sample size, or a bug in the temperature-scaling procedure is a diagnostic follow-up.

9 Conclusion

This work establishes a reproducible LODO baseline for cross-domain ordinal DR grading on four public sources, characterises the cross-architecture variance, tests a pre-registered segmentation-auxiliary-channel ablation, and reports a per-threshold calibration diagnostic that independently confirms the prior-literature finding that post-hoc temperature scaling fails under DR cross-dataset shift. The pre-registered 2pp QWK improvement threshold for the segmentation auxiliary was crossed on APTOS by the 4-ch soft variant (+2.92pp) and on EyePACS by the same variant (+2.01pp, just above), missed on DDR (+1.33pp), and decisively *not* crossed on Messidor-2 (−5.97pp); the cross-fold mean Δ QWK is +0.07pp for 4-ch soft, −1.86pp for 4-ch Tversky, and −2.13pp for 4-ch morph. The class-balancing ablation shows the unbalanced baseline is uniformly +0.92pp to +4.12pp better than the balanced baseline (mean Δ QWK +2.70pp) — the dominant single ablation effect. The per-threshold calibration diagnostic (§4) shows both per-threshold and single-global temperature scaling *increase* held-out ECE (mean Δ ECE +0.30 absolute), with the fitted global temperature saturating at $T = 5.0$ on every fold; this is an independent confirmation of the prior-literature finding (Poyrazer et al., 2026) reproduced here on a CORN ordinal regression head with the per-threshold decomposition. The honest reading is that the segmentation auxiliary channel is *not* a robust cross-domain improvement at the effect size pre-registered; the dominant negative ablation effect is the 3:1 class-balancing step; and the calibration diagnostic is the methodological contribution that survives both ablations.

10 Future Work

The analysis above is the complete set of empirical findings produced by this study. The following items are defined in the methodology or named in the results summary (`docs/results.md`) but are not yet executed or populated. Each item has a known compute budget and a known implementation surface; none requires methodological design work.

- **Bootstrap confidence intervals for all reported metrics.** *Completed 2026-08-12.* Per-sample predictions dumped to `saved/logs/per_sample_predictions/` for all five ConvNeXt-Tiny variants $\times$ four folds (20 inference runs, no retraining). 1000-resample paired bootstrap on the per-sample arrays yields the 95% confidence intervals for QWK, accuracy, and macro-F1 reported in Table 1 (per fold) and alongside the per-fold point estimates in ġ2–ġ4 and ġ6 of the results summary. JSON artifact: `saved/logs/bootstrap_cis.json`. Concrete observation: on EyePACS the segmentation gain is statistically meaningful at $\alpha = 0.05$ (4-ch soft 0.460 [0.453, 0.467] vs 3-ch balanced 0.440 [0.433, 0.448], non-overlapping CIs); on APTOS the gain is also significant (4-ch soft 0.862 [0.851, 0.872] vs 3-ch balanced 0.832 [0.822, 0.842], non-overlapping CIs). The Messidor-2 gap is significant and in the wrong direction (4-ch soft CI [0.624, 0.703] lies wholly below the 3-ch balanced CI [0.692, 0.760], with overlap on [0.692, 0.703]). The DDR and APTOS gaps are within the same neighbourhood as the EyePACS pattern.
- **Decision-boundary threshold tuning (ġ7 of the results summary).** *Completed 2026-08-12.* Post-hoc additive bias $b \in [-0.40, +0.40]$ on all four CORN sigmoid thresholds with honest 50/50 out-of-bias-sample CV (5 reps, stratified by class). Mean cross-cell Δ QWK is $+0.025$; every (variant, fold) cell except two sees non-negative Δ QWK. Largest single gain is 3-ch unbalanced $\times$ EyePACS ($+0.0897$, bias $b = -0.23$). Per-cell results in Table 7. No logits or weights are modified. Full per-(b) sweep curves in `docs/analysis/figures/decision_boundary/`.
- **Per-threshold calibration of the auxiliary-channel variants.** *Completed 2026-08-12.* Extended the per-threshold and global temperature scaling diagnostic to the 3-ch unbalanced, 4-ch soft, 4-ch Tversky, and 4-ch morph variants. Honest 50/50 5-rep CV (temperature fitted on train half, ECE on test half). JSON artifact: `saved/logs/calibration_all_variants.json`, markdown: `saved/logs/calibration_summary.md`. The global-temperature pathology ($T \rightarrow 5.0$ upper bound, mean Δ ECE $\approx +0.10$ to $+0.18$) is reproduced on every variant on every fold. The segmentation auxiliary does not rescue it.
- **Seed sensitivity for the high-priority variants (ġ9 of the results summary).** Repeat the 3-channel baseline (ConvNeXt-Tiny only) and the 4-channel soft-mask variant across a small pre-registered seed set (e.g., 3 seeds per fold). The seed set is to be pre-registered before any seed-sensitivity run is started. Estimated compute: 4 folds $\times$ 2 variants $\times$ 2 additional seeds = 16 runs at ~ 25 min each $\approx \sim 7$ hours.
- **Single-source failure mode characterization (ġ10 of the results summary).** The qualitative failure-mode characterisation is now populated for the 3-channel ConvNeXt-Tiny baseline in this analysis (ġ5, Tables 8–11). The ġ10 table of the results summary is not yet populated for the full variant matrix.
- **Full backbone family.** The architecture comparison is across three backbones (ConvNeXt-Tiny, DeiT-3 Small/384, MaxViT-Tiny/384). ConvNeXt-Small (~ 50 M parameters) and SwinV2-Base (192 $\rightarrow$ 384 transfer, ~ 88 M parameters) are not evaluated. The runner, the loss function, and the LODO split are all in place; only the model yaml and the per-arch hyperparameter

override (SwinV2’s batch size 12) need to be configured. Estimated compute: ~5–8 hours of training across 8 folds.

- **Complete the auxiliary-channel ablation matrix.** The 4-channel Sobel variant, the 5-channel (soft + Sobel) variant, and the 5-channel (Tversky + Sobel) variant are not run in this study. The Sobel auxiliary pool is already pre-computed at preprocessing time; the 5-channel variants require a single training-time concatenation change. Estimated compute: ~12–14 runs $\times$ ~25 min/run on ConvNeXt-Tiny = ~5–6 hours.
- **Extend the source pool to the remaining four GDRBench sources.** The four sources used here are a strict subset of GDRBench’s eight; the remaining four add cases for different acquisition protocols, different patient populations, and a different reference-grading scheme. The same pipeline applies without modification; the work is in the data acquisition and grading-scheme reconciliation, not in the modeling.
- **Investigate the temperature-saturation pathology.** The fitted global temperature saturates at the upper bound ($T = 5.0$) on every fold *and every variant tested* (the new all-variants diagnostic confirms it on 19/20 (variant, fold) cells), suggesting the optimizer is being aggressively pushed to extreme softening that does not transfer. Understanding why this happens — whether due to the validation-set sample size, the loss surface, or a bug in the temperature-scaling procedure — is a diagnostic follow-up before drawing strong conclusions about the post-hoc calibration failure mode.